



Abstract

Our goal is to understand how deconfinement phase-transitions effect the evolution of non-axisymmetric fluid instabilities. Studies show [3] when these non-axisymmetric instabilities develop, they affect the gravitational waves produced by the merger remnants. We run general relativistic simulations of differentially rotating stars, that have the potential to model post merger remnants. As shown by [4], quasi-toroidal NS configuration can appear as transient states after a BNS merger and these states are unstable to the one-arm spiral instability.

One-Arm Spiral Instability

The one-arm Spiral Instability develops when vortices created by the merger cause a toroidal fluid configuration where the maximum density occurs in a ring around the center-of-mass. the equatorial rest-mass density Although we initially tend to see bar modes develop, once the instability saturates, the $m = 1$ mode dominates.

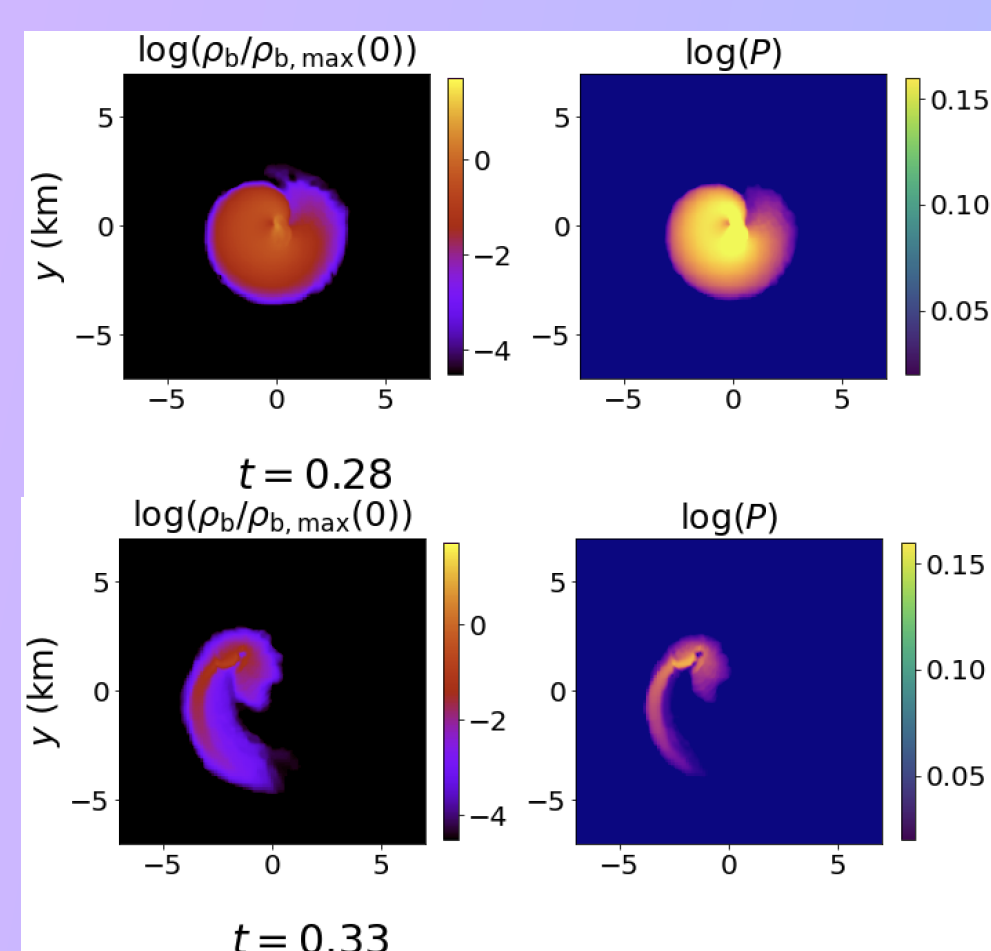
Density Perturbation

Our goal was to excite the one arm spiral instability explicitly within the simulation. In order to break angular symmetry in such a way, we initiated a density perturbation in the azimuthal direction defined by the added density term below.

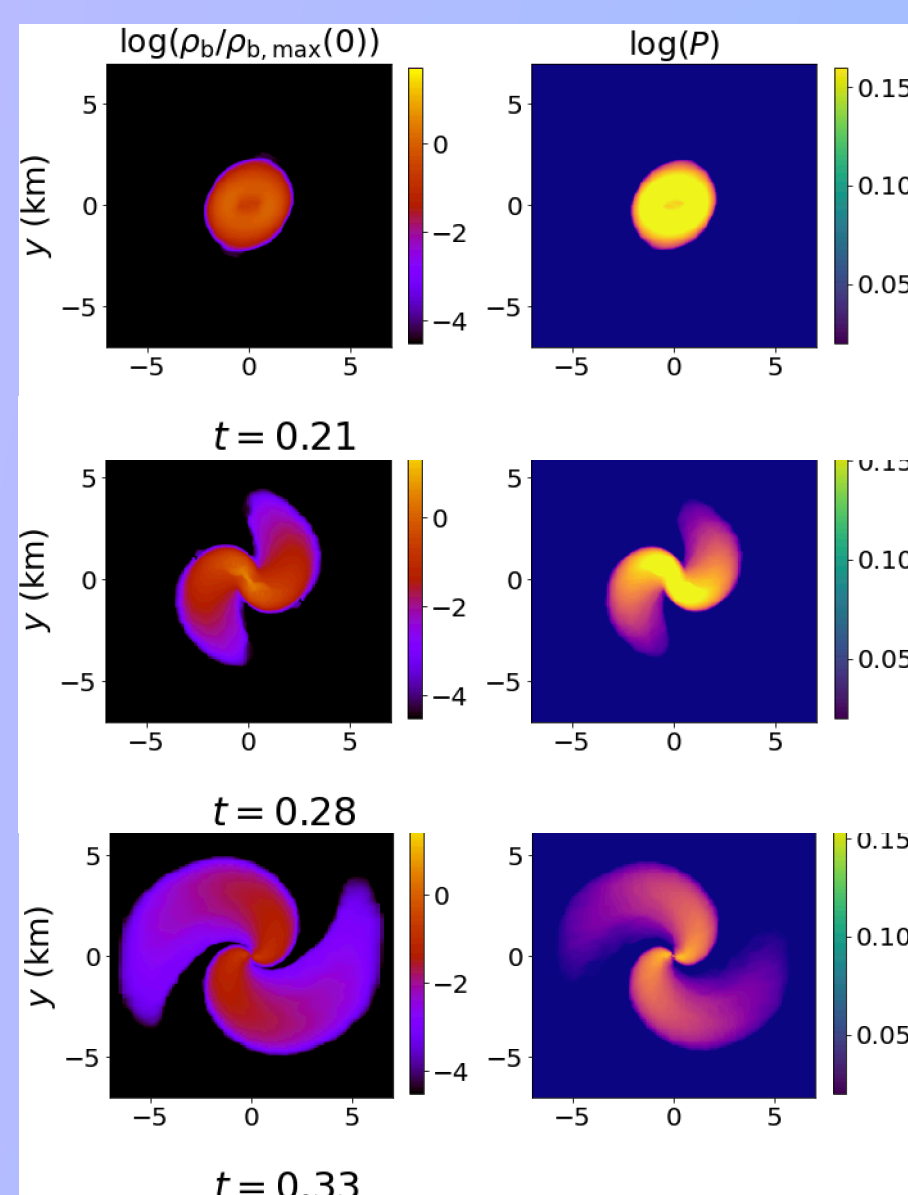
$$\rho_o \longrightarrow \rho_o \left(1 + A \frac{\omega \sin(m\phi)}{r_e} \right)$$

We ran some simulations to test our perturbation for the $m = 1$ and $m = 2$ modes. For these tests, we assumed a $\Gamma = 2$ polytrope.

m = 1

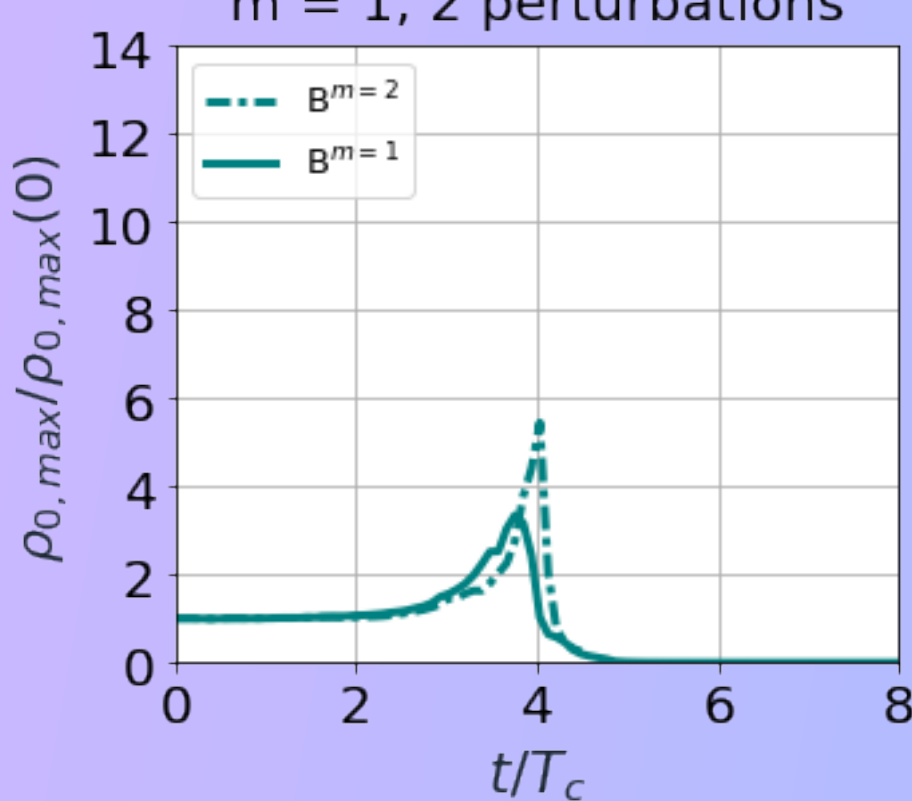


m = 2

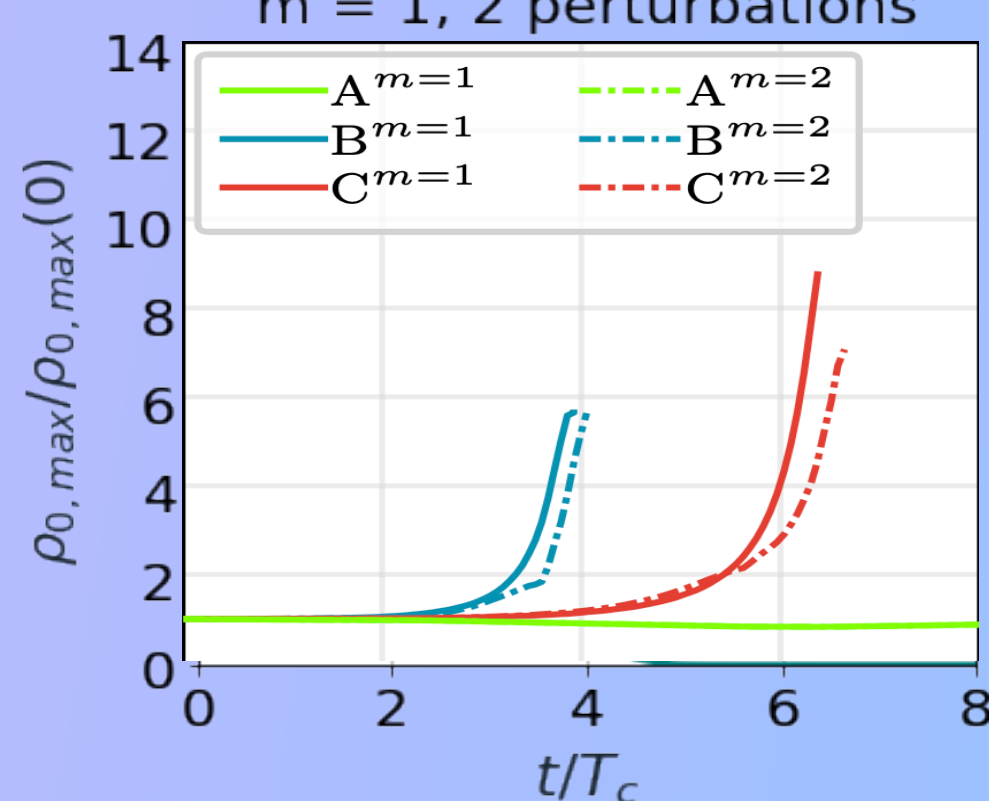


$$\omega = \sqrt{x^2 + y^2}$$

m = 1, 2 perturbations



m = 1, 2 perturbations



General Relativistic Hydrodynamic Evolution Code

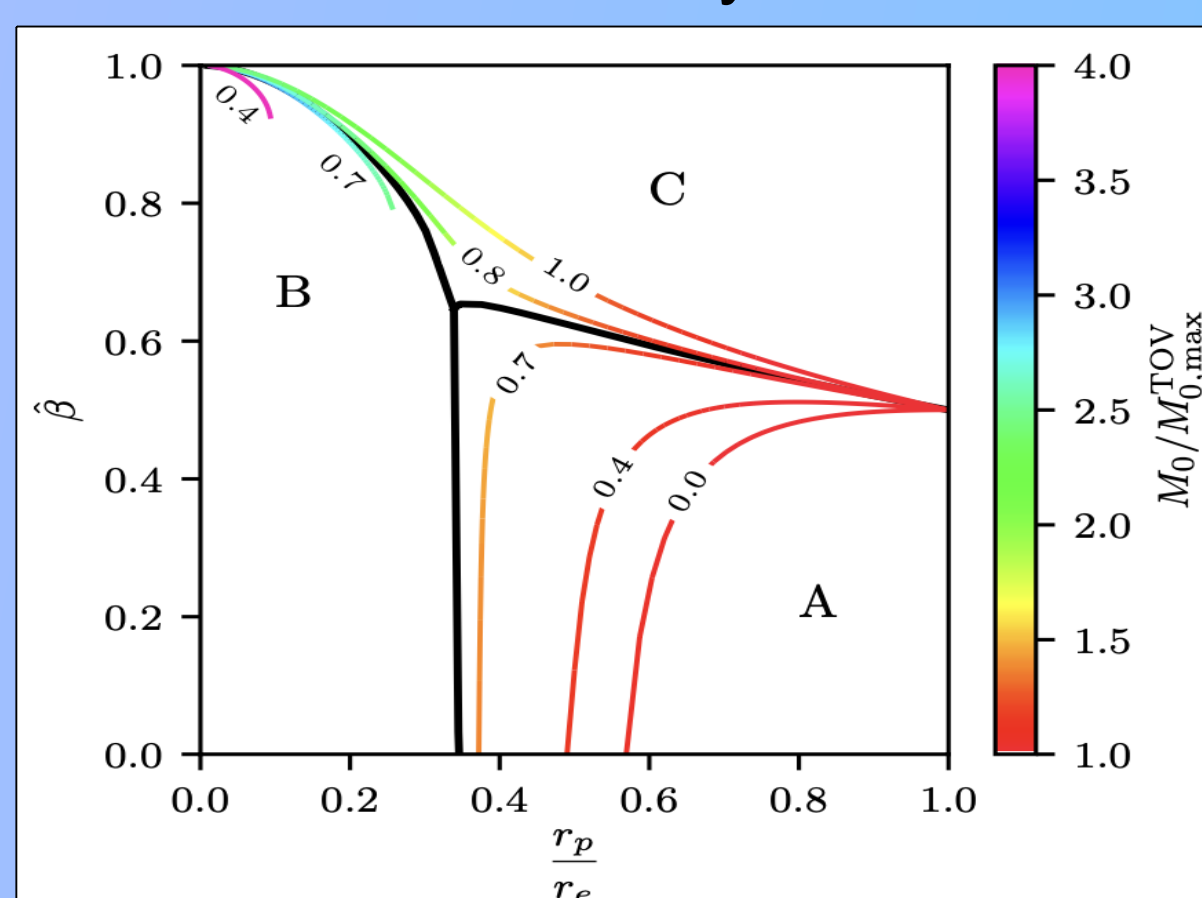
Our code evolves the stress energy tensor and the space-time metric by solving Einstein's field equations numerically for small increment in proper time for each block on a discrete grid. Our code works with the following stress-energy tensor:

$$T_{\mu\nu} = (\rho_b h + b^2) u_\mu u_\nu + (P + \frac{b^2}{2}) g_{\mu\nu}$$

The stress-energy tensor describes the matter and energy content in a system. This provides the right hand side for Einstein's equations:

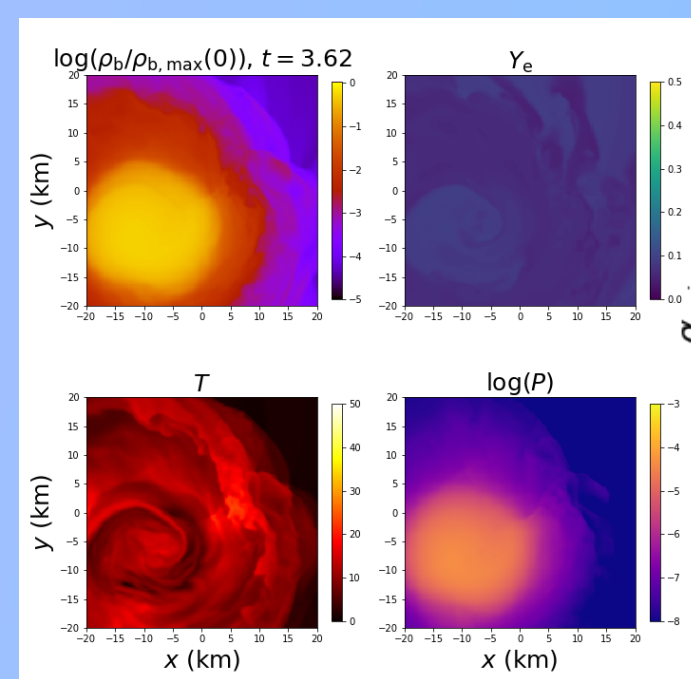
$$G_{\mu\nu} = 8\pi T_{\mu\nu}$$

We use finite-temperature EOS tables to create these models. We compare spheroidal models (max density at center) to quasi-toroidal models (max density occurs in a ring around center). To the right are the coefficients from the decomposition of rest-mass density into various instability modes.

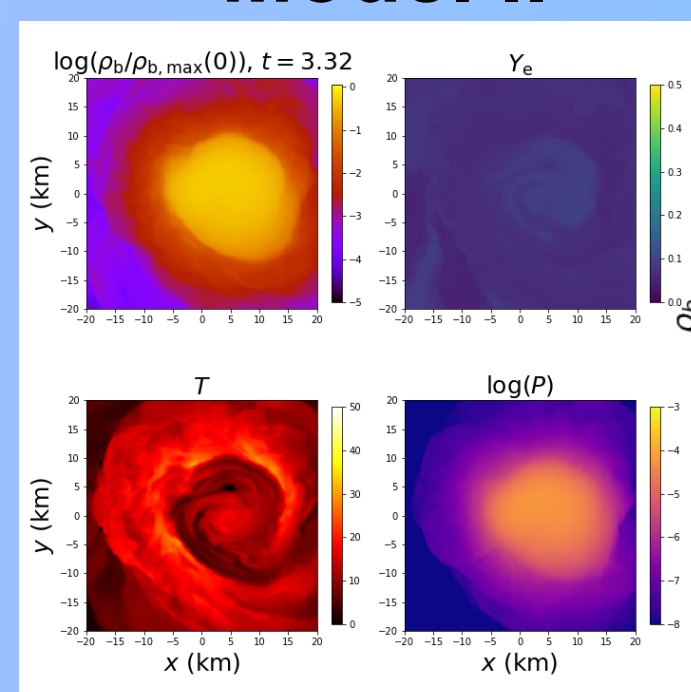


using BBKF 1.3 EOS. We consider two models with equal mass but different radii. Above to the right are a few snapshots of the various fluid variables. Both Type C sims show non-physical drifts of the C.M.

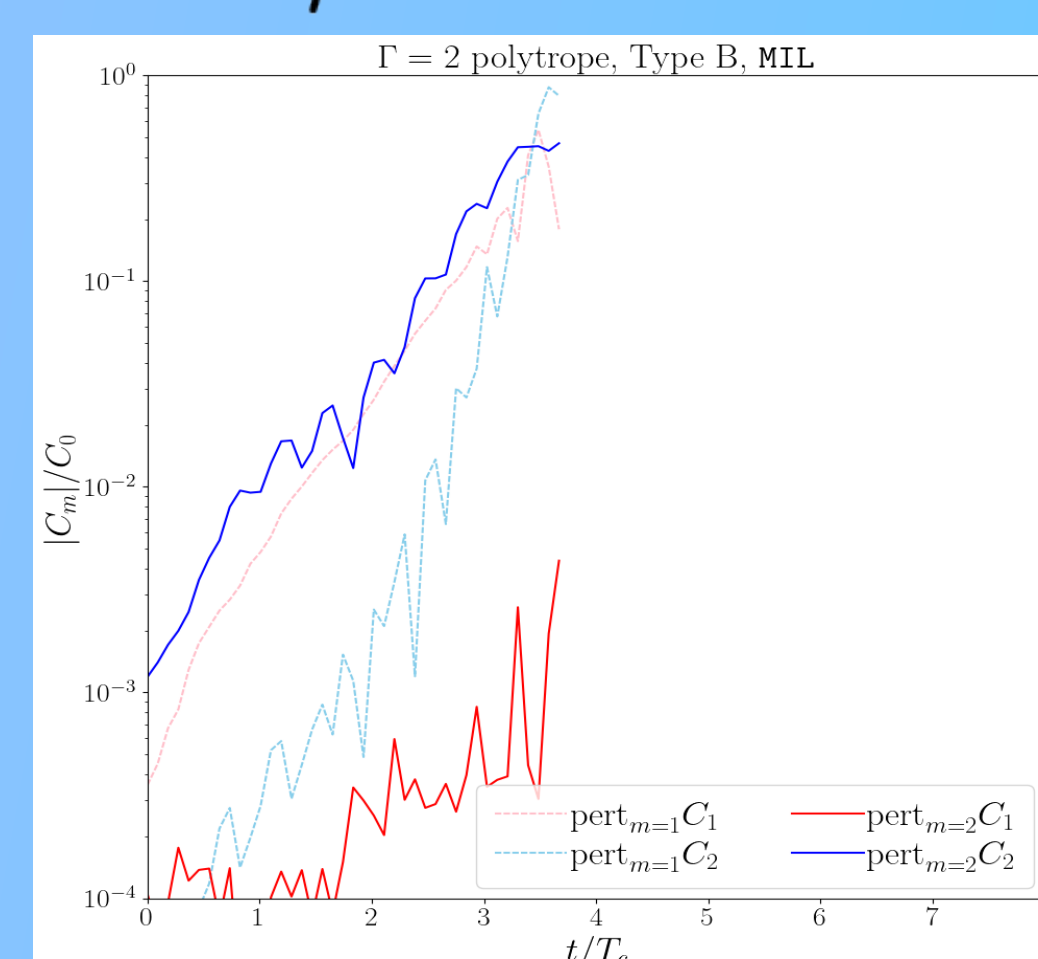
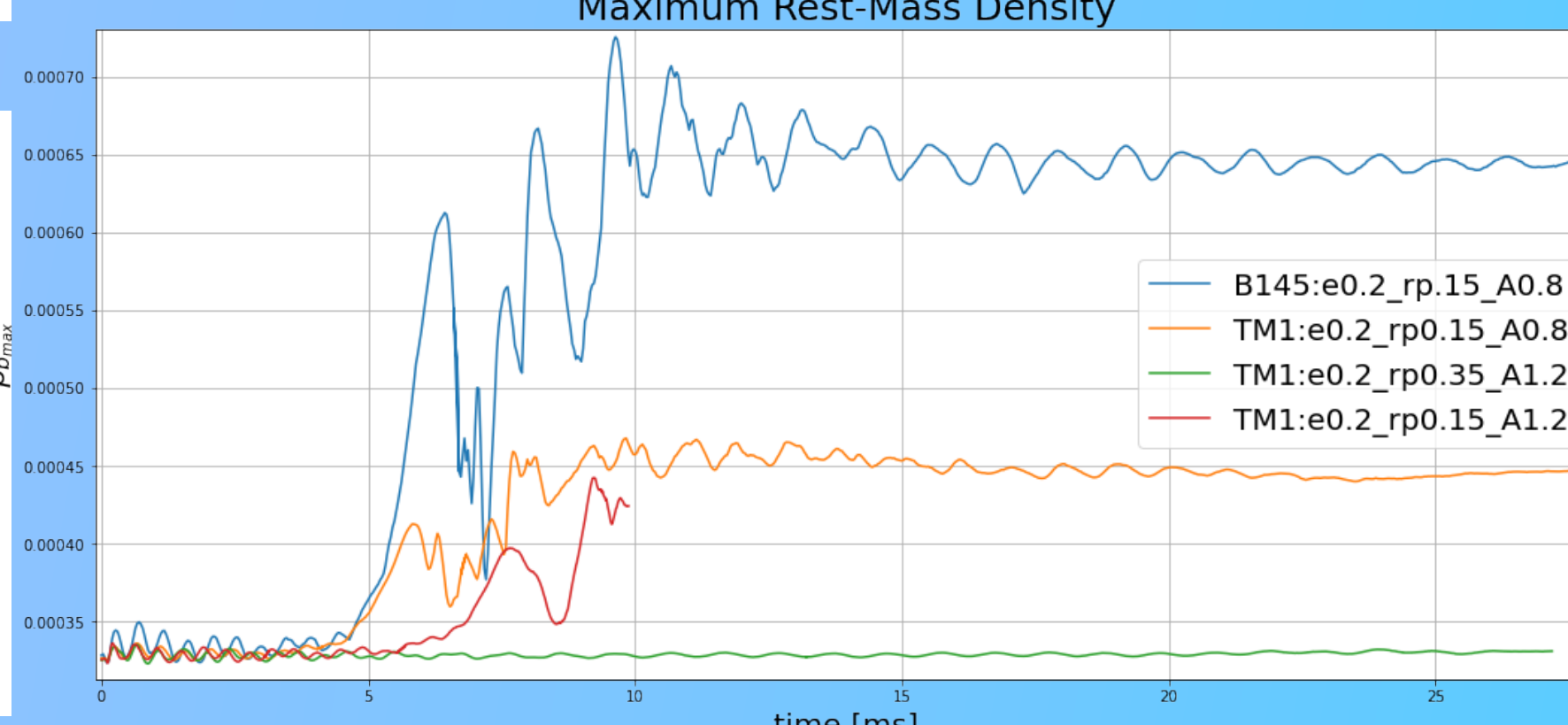
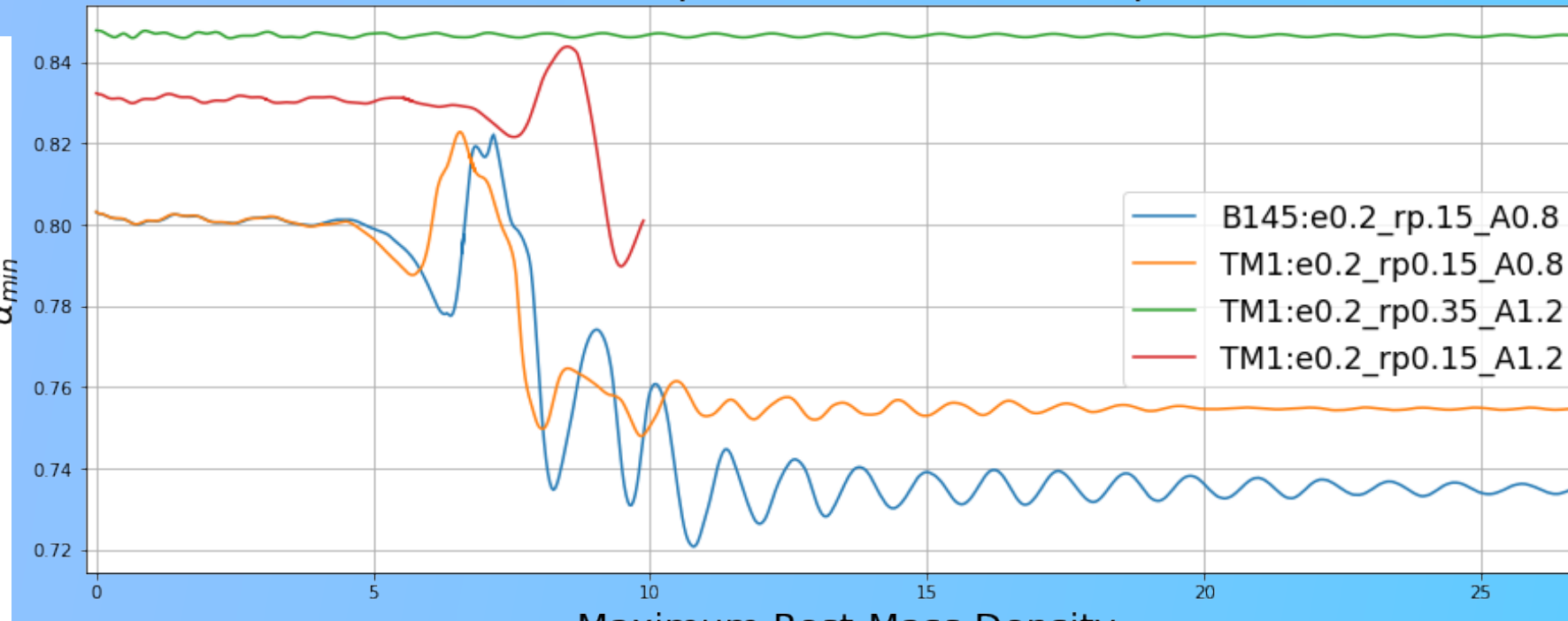
Model I



Model II



Gravitational Collapse Over Time: α Lapse Function

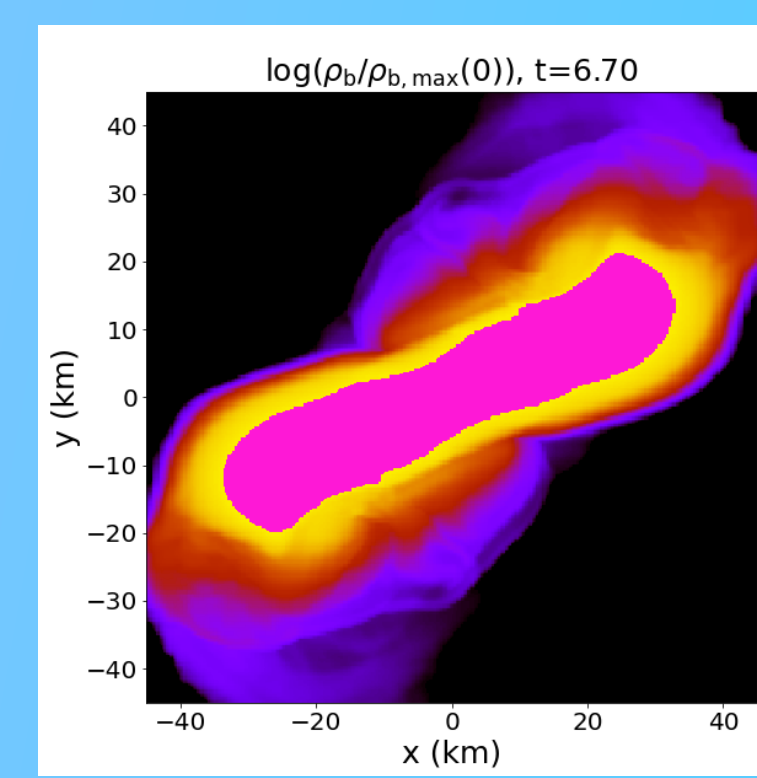


We've classified the models into types A, B, and C, whose distinctions are shown in the solution space for a $\Gamma = 2$ polytope projected on the left.

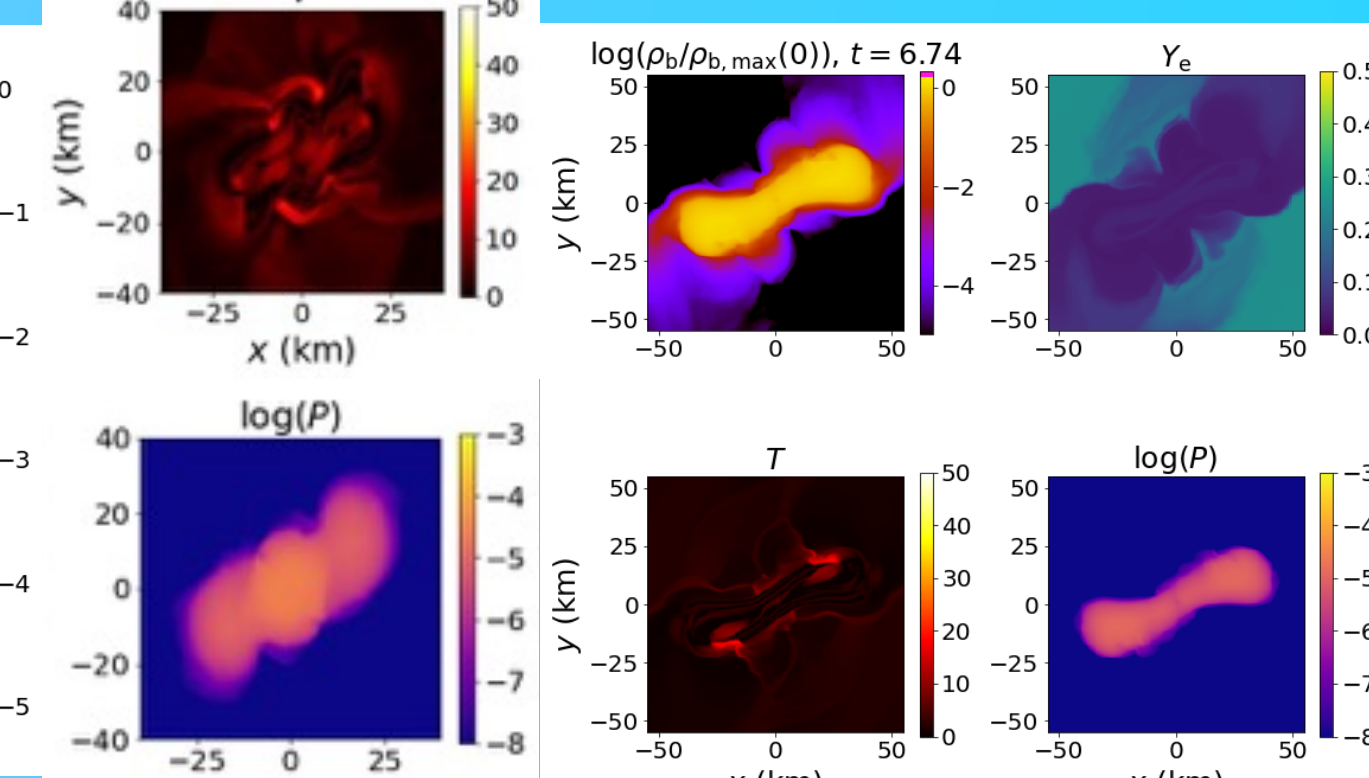
Results

We've 1st built type C models

B145-150



TM1-150-A0.8

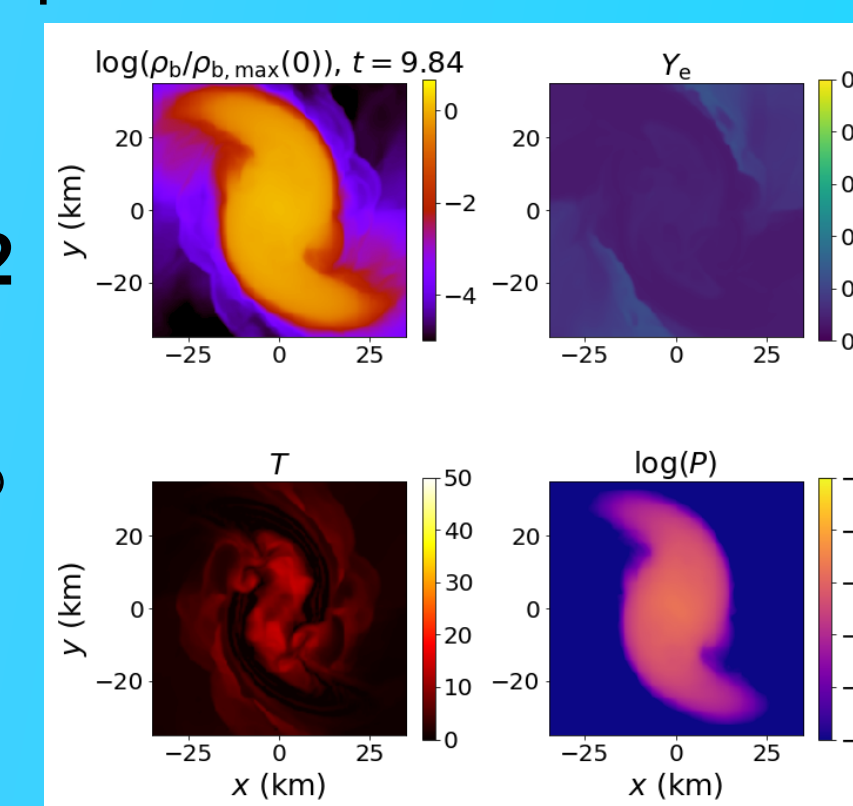
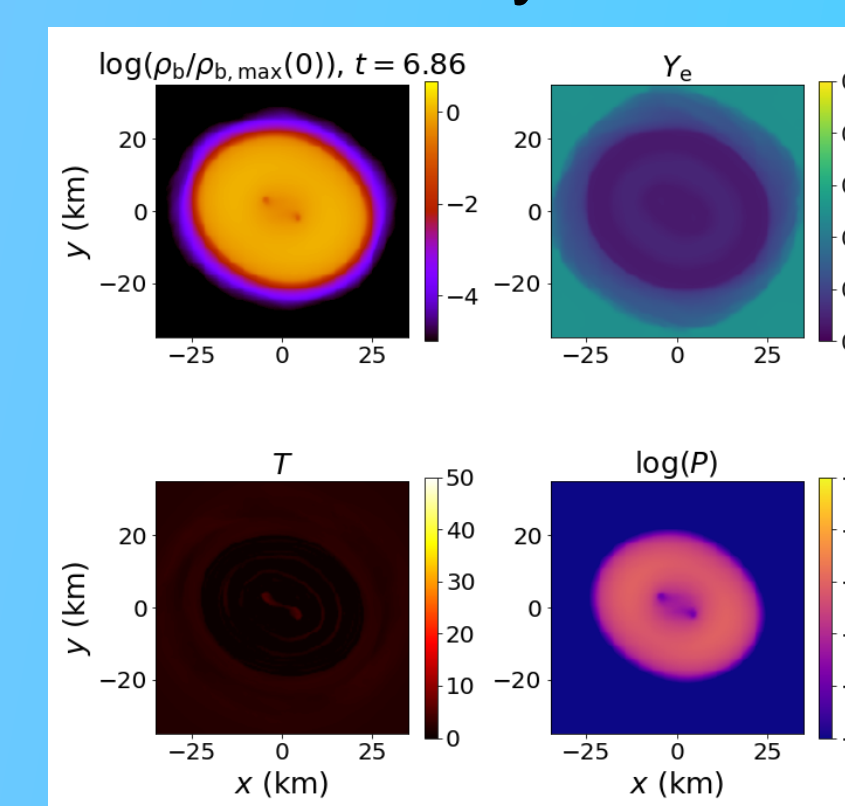


The development of the spiral instabilities were difficult to see even with the density perturbation. Models I and II were still too unstable. To solve this, we build more stable models using the B145 and TM1 EOS and doubled the perturbation amplitude. You can finally see the bar mode develop in B145 above.

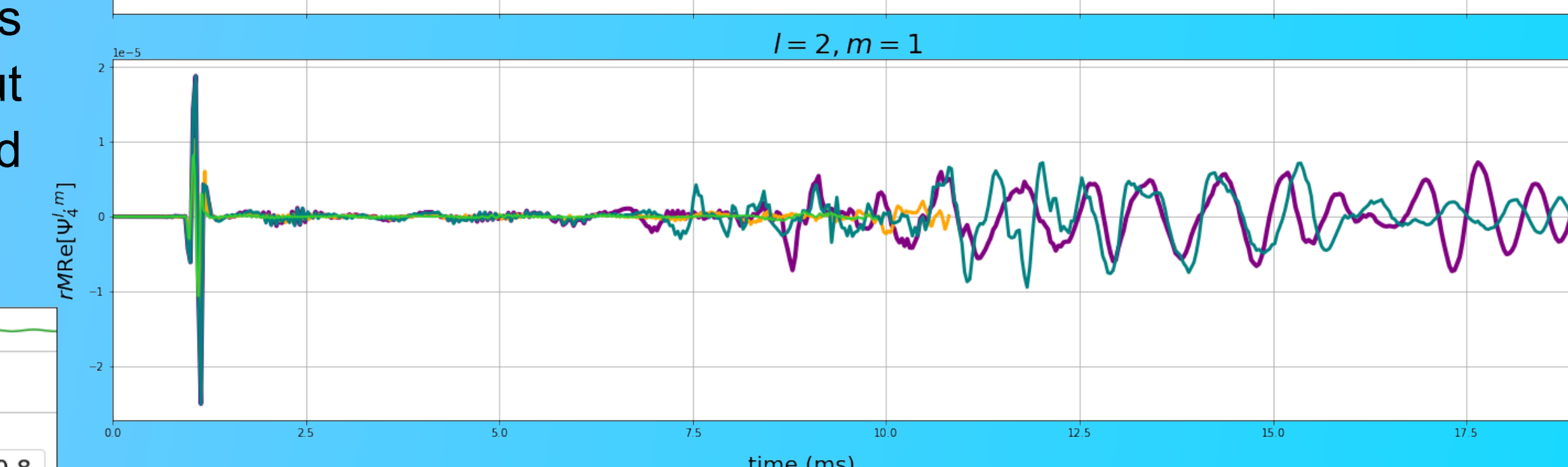
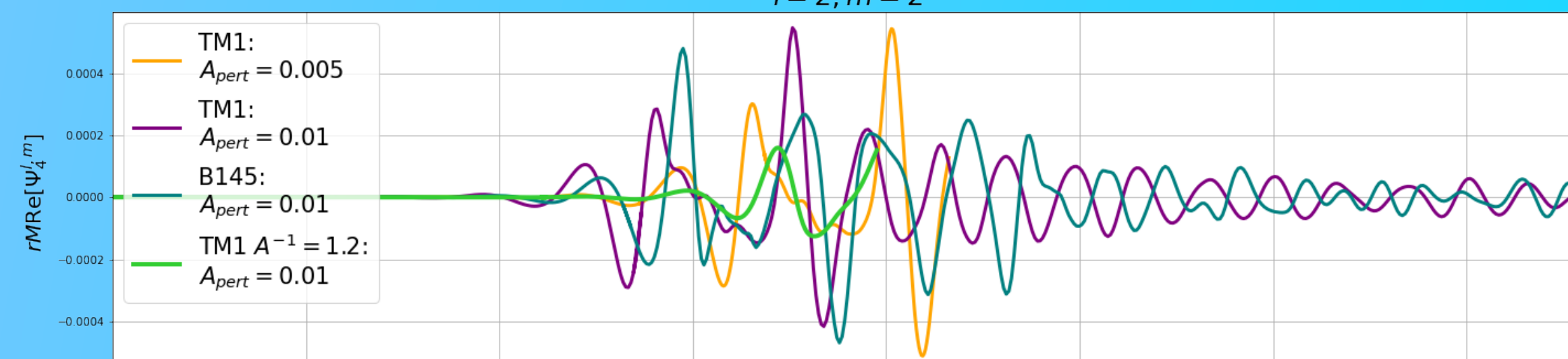
Model V TM1-150-A0.2

$M = 1.485 M_\odot$

$$\frac{T}{W} = 0.273$$



Gravitational Wave Decomposition for TM1 and B145 for Radius = 295.329 km $l = 2, m = 2$



EOS Model	Type	Mass	A^{-1}	$\frac{T}{W}$	β	$\frac{T}{W}$	E_{max} density $[g/cm^3]$	Unstable to m=1
BBKF1.3.e0.855	A	$2.7002 M_\odot$	0.536	0.370	0.59947	0.2356	$0.8550 \cdot 10^{15}$	unstable
BBKF1.3.e1.500	A	$2.9389 M_\odot$	0.450	0.390	0.61127	0.2303	$2.000 \cdot 10^{14}$	×
BBKF1.3.0.9754	C	$2.2751 M_\odot$	1.00	0.548	0.69757	0.1456	$9.7540 \cdot 10^{14}$	unstable
BBKF1.3.e1.225	C	$2.2902 M_\odot$	1.00	0.536	0.70659	0.1496	$1.2250 \cdot 10^{15}$	unstable
B145.150.A0.800	B	$1.9517 M_\odot$	0.800	0.150	0.89156	0.3080	$2.000 \cdot 10^{14}$	✓
TM1.350.A1.200	C	$0.99343 M_\odot$	1.2000	0.350	0.76720	0.2276	$2.000 \cdot 10^{14}$	×
TM1.150.A0.800	B	$1.9513 M_\odot$	0.800	0.150	0.89115	0.3081	$2.000 \cdot 10^{14}$	✓
TM1.150.A1.200	C	$1.4853 M_\odot$	1.200	0.150	0.94379	0.2725	$2.000 \cdot 10^{14}$	✓

References

- [2] W. E. East, V. Paschalidis, F. Pretorius, and S. L. Shapiro, "Relativistic simulations of eccentric binary neutron star mergers: One-arm spiral instability and effects of neutron star spin," Physical Review D, vol. 93, jan 2016.
- [3] P. L. Espino, A. Prakash, D. Radice, and D. Logoteta, "Revealing phase transition in dense matter with gravitational wave spectroscopy of binary neutron star mergers," 2023.
- [4] P. L. Espino, V. Paschalidis, T. W. Baumgarte, and S. L. Shapiro, "Dynamical stability of quasitoroidal differentially rotating neutron stars," Physical Review D, vol. 100, aug 2019.